

Relational Schema Design

```
MEMBERS(  
    member_id (primary key),  
    first_name,  
    last_name,  
    email (unique),  
    phone,  
    membership_date  
)
```

```
AUTHORS(  
    author_id (primary key),  
    first_name,  
    last_name  
)
```

```
BOOKS(  
    book_id (primary key),  
    title,  
    isbn (unique),  
    publication_year,  
    available_copies,  
    author_id FK → AUTHORS.author_id  
)
```

```
LOANS(  
    loan_id (primary key),  
    member_id FK → MEMBERS.member_id,  
    book_id FK → BOOKS.book_id,  
    checkout_date,  
    due_date,  
    return_date  
)
```

The author_id foreign key in the BOOKS relation connects each book to its author. The member_id foreign key in the LOANS relation connects each loan to the member who borrowed the book. The book_id foreign key in the LOANS relation connects each loan to the borrowed book.

The LOANS relation resolves the many-to-many relationship between members and books. A member can borrow many books over time, and a book can be borrowed by many members over time.